

Predictor Selection

Choosing which variables to include in a regression model

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Configuring R

Functions from these packages will be used throughout this document:

```
library(conflicted) # check for conflicting function definitions
# library(printr) # inserts help-file output into markdown output
library(rmarkdown) # Convert R Markdown documents into a variety of formats.
library(pander) # format tables for markdown
library(ggplot2) # graphics
library(ggfortify) # help with graphics
library(dplyr) # manipulate data
library(tibble) # `tibble`s extend `data.frame`s
library(magrittr) # `%>%` and other additional piping tools
library(haven) # import Stata files
```

```

library(knitr) # format R output for markdown
library(tidyr) # Tools to help to create tidy data
library(plotly) # interactive graphics
library(dobson) # datasets from Dobson and Barnett 2018
library(parameters) # format model output tables for markdown
library(haven) # import Stata files
library(latex2exp) # use LaTeX in R code (for figures and tables)
library(fs) # filesystem path manipulations
library(survival) # survival analysis
library(survminer) # survival analysis graphics
library(KMsurv) # datasets from Klein and Moeschberger
library(parameters) # format model output tables for
library(webshot2) # convert interactive content to static for pdf
library(forcats) # functions for categorical variables ("factors")
library(stringr) # functions for dealing with strings
library(lubridate) # functions for dealing with dates and times

```

Here are some R settings I use in this document:

```

rm(list = ls()) # delete any data that's already loaded into R

conflicts_prefer(dplyr::filter)
ggplot2::theme_set(
  ggplot2::theme_bw() +
    # ggplot2::labs(col = "") +
    ggplot2::theme(
      legend.position = "bottom",
      text = ggplot2::element_text(size = 12, family = "serif")))

knitr::opts_chunk$set(message = FALSE)
options('digits' = 6)

panderOptions("big.mark", ",")
pander::panderOptions("table.emphasize.rownames", FALSE)
pander::panderOptions("table.split.table", Inf)
conflicts_prefer(dplyr::filter) # use the `filter()` function from dplyr() by default
legend_text_size = 9
run_graphs = TRUE

```

Acknowledgements

This content is adapted from:

- Vittinghoff et al. (2012), Chapter 10
- Heinze et al. (2018)

1 Introduction

1.0.1 Why predictor selection matters

In regression modeling, we often have many potential predictors available, but including too many predictors can cause problems:

- **Overfitting:** the model fits noise in the training data and predicts poorly for new data.
- **Multicollinearity:** highly correlated predictors make coefficients unstable and hard to interpret.
- **Loss of precision:** including unnecessary predictors increases standard errors.

On the other hand, excluding important predictors can lead to:

- **Confounding bias:** omitting a common cause of the exposure and outcome distorts the estimated association.
- **Reduced power:** failing to adjust for strong predictors of the outcome can reduce statistical power.

1.0.2 Strategies for predictor selection

Vittinghoff et al. (2012, chap. 10) describe several general approaches to predictor selection:

1. **Science-driven (purposeful) selection:** Use subject-matter knowledge and a causal diagram (DAG) to identify which variables to include. This is the preferred approach when the goal is causal inference.
2. **Testing-based selection:** Use statistical tests to decide which predictors to include. Examples include forward selection, backward elimination, and stepwise selection. These methods have well-known problems and should be used with caution (see Section 3).
3. **Criterion-based selection:** Compare models using information criteria such as AIC or BIC. These criteria balance model fit against complexity (see Section 4).
4. **Penalized regression:** Use regularization methods (such as the LASSO) that shrink coefficients toward zero, effectively performing selection and estimation simultaneously (see Section 5).

1.0.3 The role of subject-matter knowledge

Regardless of which selection strategy is used, subject-matter knowledge should guide model building.

A regression model can only tell us about *associations* — whether certain predictors are correlated with the outcome in our data. It cannot, on its own, tell us whether those associations reflect causal effects.

To draw causal conclusions, we need:

1. A causal model (such as a DAG) specifying assumed causal relationships.
2. Appropriate adjustment for confounders.
3. Careful avoidance of adjusting for mediators or colliders.

These concepts are discussed in Section 2.0.1.

2 Causal Thinking and DAGs

2.0.1 Directed Acyclic Graphs (DAGs)

A **Directed Acyclic Graph (DAG)** is a graphical model that encodes assumed causal relationships among variables.

In a DAG:

- **Nodes** represent variables.
- **Directed edges** (arrows) represent direct causal effects.
- The graph is **acyclic**: there are no directed cycles (a variable cannot cause itself, even indirectly).

DAGs are used in causal inference to:

- Identify **confounders** that must be adjusted for.
- Identify **mediators** that should generally *not* be adjusted for when estimating the total causal effect.
- Identify **colliders** that should *not* be adjusted for (adjusting for a collider opens a non-causal path).

2.0.2 Key concepts from DAG theory

Definition 2.1 (Confounder). A **confounder** of the relationship between an exposure X and outcome Y is a variable C that:

1. Is associated with the exposure X .
2. Is a cause of the outcome Y .
3. Is not on the causal pathway from X to Y .

Adjusting for confounders removes confounding bias.

Definition 2.2 (Mediator). A **mediator** M is a variable that lies on the causal pathway from the exposure X to the outcome Y :

$$X \rightarrow M \rightarrow Y$$

Adjusting for a mediator blocks the indirect causal pathway through M , so the estimated coefficient for X reflects only the *direct* effect of X on Y not through M .

Whether to adjust for mediators depends on the research question:

- If you want the **total causal effect** of X on Y , do **not** adjust for M .
- If you want the **direct effect** of X on Y not through M , adjust for M .

Definition 2.3 (Collider). A **collider** is a variable L that is caused by two or more variables on a path:

$$X \rightarrow L \leftarrow Y$$

Adjusting for a collider opens a non-causal (backdoor) path between X and Y , introducing **collider bias**.

2.0.3 HERS study DAG

The “heart and estrogen/progestin study” (HERS) was a clinical trial of hormone therapy for prevention of recurrent heart attacks and death among 2,763 post-menopausal women with existing coronary heart disease (CHD) (Hulley et al. 1998).

Figure 1 shows a DAG for the HERS study, representing the assumed causal structure among key variables.

The main causal question is:

What is the effect of hormone therapy (HT) on recurrent CHD events?

```
library(dagitty)

hers_dag <- dagitty(
  'dag {
    HT [exposure, pos="0,0"]
    CHD [outcome, pos="4,0"]
    age [pos="2,2"]
    smoking [pos="-1,-2"]
    diabetes [pos="1,-1.5"]
    BMI [pos="-1,1"]
    LDL [pos="2,1"]
    HDL [pos="2,-1"]
    statins [pos="3,2"]
    SBP [pos="3,-2"]

    HT -> CHD
    HT -> HDL
```

```

    HT -> LDL
    age -> CHD
    age -> HT
    smoking -> CHD
    smoking -> HDL
    BMI -> CHD
    BMI -> diabetes
    BMI -> SBP
    BMI -> LDL
    diabetes -> CHD
    diabetes -> LDL
    LDL -> CHD
    HDL -> CHD
    statins -> LDL
    SBP -> CHD
  }'
)
plot(hers_dag)

```

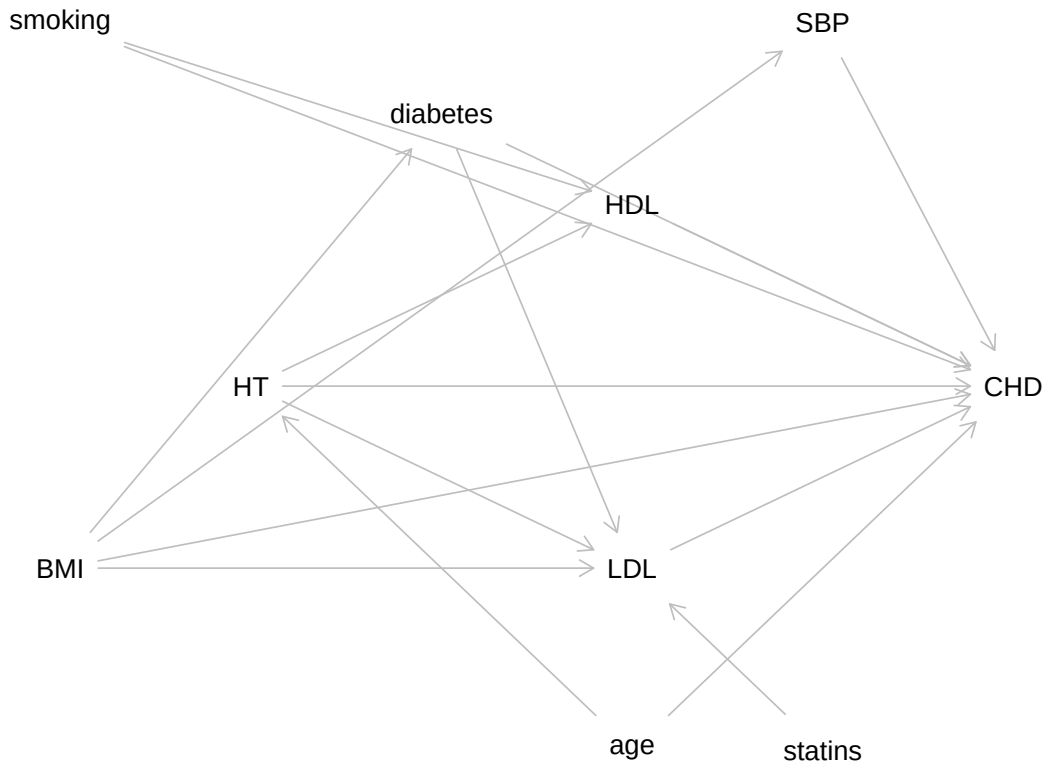


Figure 1: Directed Acyclic Graph (DAG) for the HERS study. HT = hormone therapy (exposure); CHD = recurrent coronary heart disease events (outcome); HDL = high-density lipoprotein; LDL = low-density lipoprotein; SBP = systolic blood pressure.

In Figure 1:

- **age** is a confounder: older women were more likely to have CHD events and may differ in their hormone therapy assignment.
- **LDL** and **HDL** are *mediators*: hormone therapy affects lipid levels, which in turn affect CHD risk.
- **BMI** is a common cause of diabetes, SBP, and LDL.

Table 1

```
adjustmentSets(
  hers_dag,
  exposure = "HT",
  outcome = "CHD",
  type = "minimal"
)
#> { age }
```

- **statins** (statin use) affects LDL independently of HT (statins are medications prescribed to lower LDL).
- This DAG is simplified; a more complete model might include additional variables such as anti-hypertensive medication use.

2.0.4 What variables should we adjust for?

Using DAG theory, we can identify the minimum adjustment set needed to estimate the total causal effect of HT on CHD.

According to the DAG in Figure 1, adjusting for **age** alone is sufficient to close all backdoor paths from HT to CHD.

The variables LDL and HDL are mediators on the path $HT \rightarrow LDL/HDL \rightarrow CHD$, so they should **not** be included when estimating the *total* causal effect of HT on CHD.

Including LDL and HDL in the model would block a part of the causal effect of HT, giving us only the *direct* effect of HT on CHD (not through lipids).

2.0.5 Analyzing LDL in the HERS study

The HERS baseline dataset includes LDL and HDL cholesterol, allowing us to examine the effect of hormone therapy on lipid levels.

Here we build models to predict LDL cholesterol, illustrating how DAG thinking guides predictor selection.

```
library(haven)
library(dplyr)

hers <- read_stata(fs::path_package("rme", "extdata/hersdata.dta"))

hers_ldl <-
  hers |>
  filter(!is.na(LDL)) |>
  mutate(
    HT = factor(HT, levels = c(0, 1), labels = c("Placebo", "HT")),
    statins = factor(statins, levels = c(0, 1), labels = c("No", "Yes")),
    diabetes = factor(diabetes, levels = c(0, 1), labels = c("No", "Yes")),
    smoking = factor(smoking, levels = c(0, 1), labels = c("No", "Yes"))
  )
```

```
library(gtsummary)

hers_ldl |>
  select(LDL, HT, age, BMI, diabetes, smoking, statins) |>
  tbl_summary(by = HT) |>
  add_overall() |>
  bold_labels()
```

Table 2: Summary table for LDL analysis variables in HERS

Characteristic	Overall N = 2,752 [†]	Placebo N = 1,379 [†]	HT N = 1,373 [†]
LDL cholesterol (mg/dl)	141 (120, 166)	141 (119, 165)	141 (120, 167)
age in years	67 (62, 72)	68 (62, 72)	67 (62, 72)
BMI (kg/m ²)	27.7 (24.6, 31.7)	27.6 (24.5, 31.7)	27.9 (24.7, 31.8)
Unknown	5	4	1
diabetes	727 (26%)	350 (25%)	377 (27%)
smoking	355 (13%)	180 (13%)	175 (13%)
statins	998 (36%)	516 (37%)	482 (35%)

[†] Median (Q1, Q3); n (%)

Table 3: Unadjusted linear regression for LDL in HERS

Characteristic	Beta	95% CI	p-value
HT			
Placebo	—	—	
HT	0.36	-2.5, 3.2	0.8

Abbreviation: CI = Confidence Interval

Unadjusted model

```
model_ldl_unadj <-
  lm(LDL ~ HT, data = hers_ldl)

model_ldl_unadj |>
  tbl_regression() |>
  bold_labels()
```

Adjusted model

Based on the DAG in Figure 1, BMI is a common cause of LDL and other outcomes (but age does not confound the HT → LDL relationship, since HT was randomized).

In a randomized trial, treatment assignment is independent of pre-treatment covariates, so confounding adjustment is not required to estimate the causal effect. However, adjusting for strong predictors of the outcome can improve precision.

```
model_ldl_adj <-
  lm(LDL ~ HT + age + BMI + diabetes + statins, data = hers_ldl)

model_ldl_adj |>
  tbl_regression() |>
  bold_labels()
```

Note that **statins** is a strong predictor of LDL (statin medications are prescribed specifically to lower LDL). Including **statins** substantially reduces residual variability, improving precision of the HT coefficient.

3 Testing-based selection

Testing-based selection methods use statistical tests (such as F-tests or likelihood ratio tests) to decide which predictors to include in a model.

Table 4: Adjusted linear regression for LDL in HERS

Characteristic	Beta	95% CI	p-value
HT			
Placebo	—	—	
HT	-0.14	-2.9, 2.6	>0.9
age in years	-0.23	-0.44, -0.02	0.035
BMI (kg/m ²)	0.46	0.19, 0.72	<0.001
diabetes			
No	—	—	
Yes	-4.6	-7.9, -1.4	0.005
statins			
No	—	—	
Yes	-17	-20, -14	<0.001

Abbreviation: CI = Confidence Interval

3.1 Stepwise regression

⚠ Caution about stepwise selection

Stepwise regression has well-known problems:

- It tends to select too many variables (overfitting).
- P-values and confidence intervals are biased after selection.
- It ignores model uncertainty.
- Results can be unstable across different samples.

Consider using subject-matter knowledge (DAGs), cross-validation, or penalized methods (such as LASSO) instead. See Heinze et al. (2018) for a thorough review.

The basic idea of stepwise selection:

- **Forward selection:** Start with no predictors; add the predictor that most improves model fit at each step.
- **Backward elimination:** Start with all predictors; remove the predictor that least worsens model fit at each step.
- **Bidirectional stepwise:** Combine forward and backward steps.

3.2 Example: stepwise selection for LDL

```
# Full model with all candidate predictors
full_model_ldl <- lm(
  LDL ~ HT + age + BMI + diabetes + smoking + statins + SBP,
  data = hers_ldl
)

# Backward elimination using AIC
step_model_ldl <- step(
  full_model_ldl,
  direction = "backward",
  trace = 0
)

summary(step_model_ldl)
#>
```

```

#> Call:
#> lm(formula = LDL ~ age + BMI + diabetes + statins + SBP, data = hers_ldl)
#>
#> Residuals:
#>      Min       1Q   Median       3Q      Max
#> -103.69  -24.08   -3.45   19.33  241.89
#>
#> Coefficients:
#>              Estimate Std. Error t value Pr(>|t|)
#> (Intercept) 142.6424     9.2977   15.34  <2e-16 ***
#> age         -0.2881     0.1087   -2.65   0.0081 **
#> BMI          0.4349     0.1337    3.25   0.0012 **
#> diabetesYes -5.3946     1.6742   -3.22   0.0013 **
#> statinsYes  -16.6397     1.4603  -11.40  <2e-16 ***
#> SBP          0.1232     0.0379    3.25   0.0012 **
#> ---
#> Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#>
#> Residual standard error: 36.8 on 2741 degrees of freedom
#> (5 observations deleted due to missingness)
#> Multiple R-squared:  0.0565, Adjusted R-squared:  0.0548
#> F-statistic: 32.8 on 5 and 2741 DF,  p-value: <2e-16

```

4 Criterion-based selection

4.1 AIC and BIC

When comparing non-nested models, we can use information criteria:

- **AIC** (Akaike Information Criterion):

$$\text{AIC} = -2\hat{\ell} + 2p$$

- **BIC** (Bayesian Information Criterion):

$$\text{BIC} = -2\hat{\ell} + p \log n$$

where $\hat{\ell}$ is the maximized log-likelihood, p is the number of model parameters, and n is the sample size.

Lower values of AIC or BIC indicate a better model. BIC applies a larger penalty for model complexity, especially for large n .

4.2 Comparing models for LDL

5 Penalized regression

When there are many candidate predictors, penalized regression methods simultaneously estimate model coefficients and perform variable selection by shrinking small coefficients toward zero.

5.1 The LASSO

The **LASSO** (Least Absolute Shrinkage and Selection Operator) adds an ℓ_1 penalty to the log-likelihood:

$$\hat{\beta}_\lambda = \arg \max_{\beta} \left\{ \hat{\ell}(\beta) - \lambda \sum_{j=1}^p |\beta_j| \right\}$$

Table 5: Comparison of LDL models by AIC and BIC

```
library(tibble)

models_ldl <- list(
  "HT only" = lm(LDL ~ HT, data = hers_ldl),
  "HT + age" = lm(LDL ~ HT + age, data = hers_ldl),
  "HT + statins" = lm(LDL ~ HT + statins, data = hers_ldl),
  "HT + age + BMI + diabetes + statins" = lm(
    LDL ~ HT + age + BMI + diabetes + statins,
    data = hers_ldl
  ),
  "Full model" = lm(
    LDL ~ HT + age + BMI + diabetes + smoking + statins + SBP,
    data = hers_ldl
  )
)

tibble(
  Model = names(models_ldl),
  AIC = sapply(models_ldl, AIC),
  BIC = sapply(models_ldl, BIC),
  Adj_R2 = sapply(
    models_ldl,
    function(m) summary(m)$adj.r.squared
  )
)

#> # A tibble: 5 x 4
#>   Model                AIC    BIC    Adj_R2
#>   <chr>             <dbl> <dbl>    <dbl>
#> 1 HT only          27807. 27825. -0.000341
#> 2 HT + age         27802. 27826.  0.00202
#> 3 HT + statins      27681. 27704.  0.0451
#> 4 HT + age + BMI + diabetes + statins 27617. 27658.  0.0511
#> 5 Full model        27609. 27662.  0.0545
```

The tuning parameter λ controls the strength of the penalty:

- $\lambda = 0$: no penalty (ordinary least squares).
- $\lambda \rightarrow \infty$: all coefficients shrink to zero.

For intermediate values of λ , some coefficients are exactly zero (excluded), while others are shrunk but non-zero.

The optimal λ is usually chosen by cross-validation.

5.2 Example: LASSO for LDL

```
library(glmnet)
library(ggfortify)

hers_ldl_complete <-
  hers_ldl |>
  select(LDL, HT, age, BMI, diabetes, smoking, statins, SBP) |>
  tidyr::drop_na()

y_ldl <- hers_ldl_complete$LDL
x_ldl <-
  hers_ldl_complete |>
  select(-LDL) |>
  mutate(
    HT = as.integer(HT == "HT"),
    statins = as.integer(statins == "Yes"),
    diabetes = as.integer(diabetes == "Yes"),
    smoking = as.integer(smoking == "Yes")
  ) |>
  as.matrix()

lasso_fit <- glmnet(x_ldl, y_ldl)
autoplot(lasso_fit, xvar = "lambda") +
  theme_bw()
```

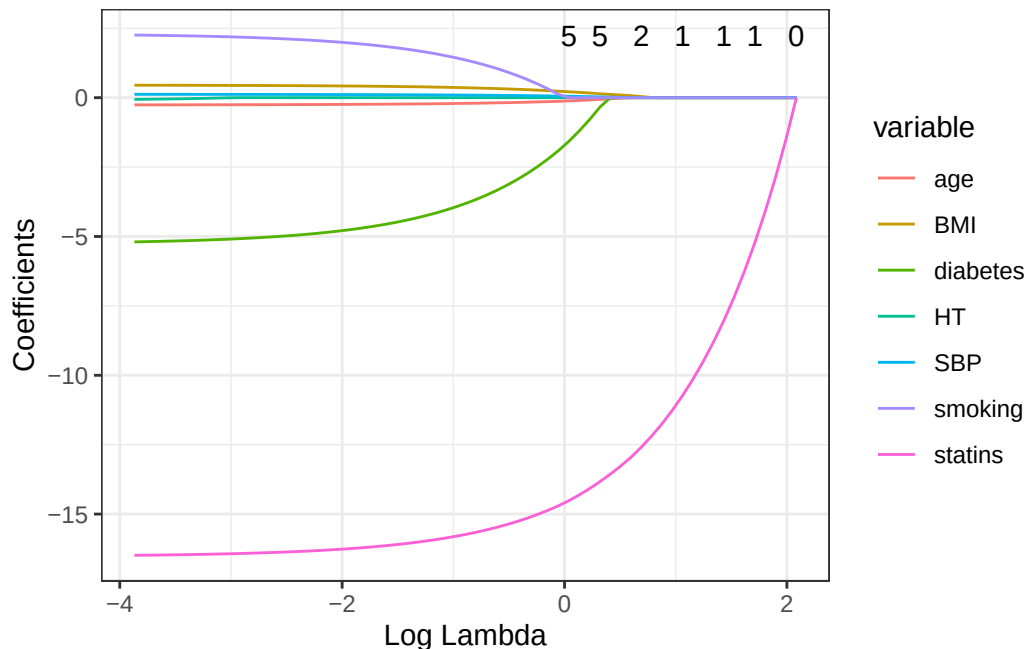


Figure 2: LASSO coefficient paths for LDL predictors in HERS

Table 6: LASSO coefficients at two values of the regularization parameter: `lambda.min` minimizes cross-validation error; `lambda.1se` uses a more conservative (larger) `lambda`, within one standard error of the minimum, producing a sparser model.

```
data.frame(
  lambda.min = as.vector(coef(cv_lasso, s = "lambda.min")),
  lambda.1se = as.vector(coef(cv_lasso, s = "lambda.1se")),
  row.names = rownames(coef(cv_lasso))
)
#> # A tibble: 8 x 2
#>   lambda.min lambda.1se
#>   <dbl>      <dbl>
#> 1    141.        146.
#> 2     0          0
#> 3   -0.235        0
#> 4    0.408        0
#> 5   -4.58        0
#> 6    1.86        0
#> 7   -16.1       -1.49
#> 8    0.108        0
```

```
cv_lasso <- cv.glmnet(x_ldl, y_ldl)
plot(cv_lasso)
```

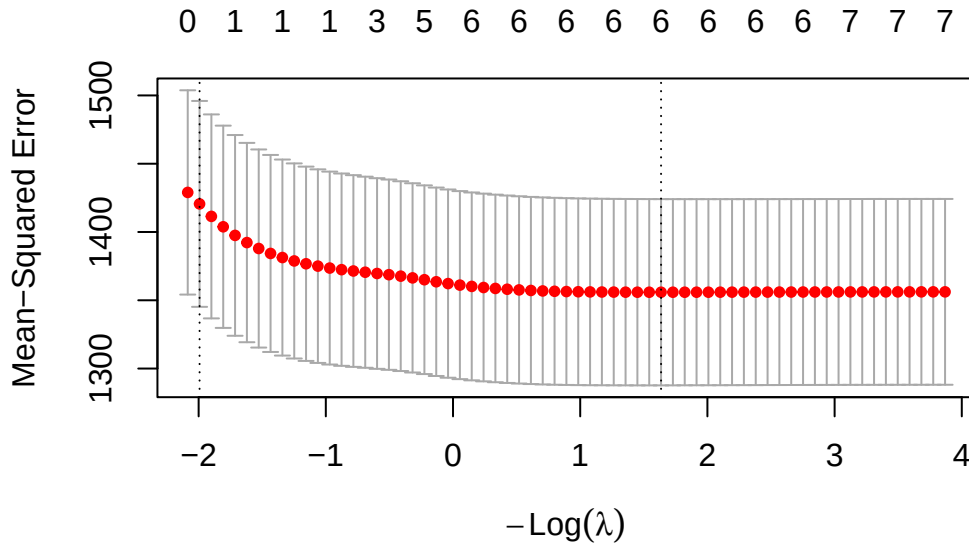


Figure 3: Cross-validated LASSO for LDL: MSE vs. $\log(\lambda)$

The LASSO identifies **statins** and **HT** as the strongest predictors of LDL cholesterol in the HERS data, consistent with the biological knowledge encoded in the DAG in Figure 1.

6 Summary

Key principles for predictor selection:

1. **Start with subject-matter knowledge.** Draw a DAG to identify confounders, mediators, and colliders.
2. **Adjust for confounders** to remove bias in causal effect estimates.
3. **Do not adjust for mediators** if you want the total causal effect.
4. **Do not adjust for colliders** to avoid introducing bias.
5. **Use AIC/BIC** to compare models of similar types.
6. **Be cautious with stepwise selection.** It tends to overfit and produces biased inference.
7. **Consider penalized regression** (LASSO, ridge) when there are many candidate predictors.
8. **In randomized trials** (like HERS), confounding adjustment is not required for causal inference, but adjusting for strong predictors of the outcome can improve precision.

Heinze, Georg, Christine Wallisch, and Daniela Dunkler. 2018. “Variable Selection – A Review and Recommendations for the Practicing Statistician.” *Biometrical Journal* 60 (3): 431–49. <https://doi.org/10.1002/bimj.201700067>.

Hulley, Stephen, Deborah Grady, Trudy Bush, et al. 1998. “Randomized Trial of Estrogen Plus Progestin for Secondary Prevention of Coronary Heart Disease in Postmenopausal Women.” *JAMA : The Journal of the American Medical Association* (Chicago, IL) 280 (7): 605–13.

Vittinghoff, Eric, David V Glidden, Stephen C Shiboski, and Charles E McCulloch. 2012. *Regression Methods in Biostatistics: Linear, Logistic, Survival, and Repeated Measures Models*. 2nd ed. Springer. <https://doi.org/10.1007/978-1-4614-1353-0>.